Chapter 9
XVA:
Valuation Adjustments in Practice

Chapters 7 and 8 arrived, by different routes, at the same uncomfortable conclusion. A swap's value depends on the curve used to discount it, and the curve depends on the collateral agreement. Where collateral does not fully cover the exposure (under a one-way CSA, a threshold, or no CSA at all), the dealer funds the position itself; where initial margin is required, that too must be funded for the life of the trade. Behind all of it stands the possibility that the counterparty defaults. The phrase "the value of the swap" therefore does not identify a number: the same cash flows are worth different amounts to different holders, under different agreements, funded differently. The adjustments that convert a textbook valuation into a price someone will actually trade at are collectively called **XVA**, and this chapter is about what they are, who charges them, and where they overlap.

Default needs a precise meaning, because every adjustment prices it. A counterparty **defaults** when it fails to meet its obligations on any of its contracts, or becomes bankrupt or insolvent. Default belongs to the counterparty, not to a trade, and it reaches all of its liabilities at once: cross-default and acceleration clauses, where a contract carries them, put its other obligations into default when one is missed, and bankruptcy stays them all together. Creditors differ only in how much each recovers, which depends on seniority and collateral. A swap that is performing perfectly is therefore caught when the counterparty defaults on a bond, a loan or another dealer. For derivatives, the ISDA Master Agreement sets the consequences. An **event of default** (among other things, a failure to pay or deliver under the agreement that is not cured within a short grace period, bankruptcy or insolvency, or, where the parties have agreed it, a default on other borrowing above a threshold amount) entitles the other party to terminate every trade under the agreement and replace them with a single net close-out amount (Section 1.5). That amount, less the collateral held, is what the counterparty may never pay. Default is a legal event, not a market price: a downgrade or a wider credit spread is not a default, although a rating trigger in the CSA can turn one into a collateral call (Section 8.1). It also differs from two neighboring definitions. Bank capital rules treat an obligor as defaulted when it is more than 90 days past due on a material obligation or is judged unlikely to pay, and a credit default swap pays on its own list of credit events (Chapter 25).

Section 9.1 defines exposure, the quantity every adjustment is built on. Sections 9.2 to 9.7 take the adjustments one at a time (credit, debit, funding, margin, collateral, and capital), and Section 9.8 adds them up on one swap. Section 9.9 treats wrong-way risk and Section 9.10 the desk that manages all of it. Section 9.11 confronts the fact that the same trade now carries three defensible values, and Section 9.12 asks what that does to the idea of a price, where the framework falls short, and where it came from.

Two warnings are needed at the outset. First, XVA is not a refinement: on a long-dated uncollateralized trade the adjustments can exceed the bid–ask spread by more than an order of magnitude and dominate the economics of the trade. Second, this is the least settled material

in the book: the components are not agreed, some may double-count, and one of them, the subject of Section 9.4, has been argued to have no business existing at all. The disputes are set out rather than resolved.

9.1 EXPOSURE

An **exposure** is the amount a party would stand to lose if its counterparty failed at a given moment, before any recovery. It is not the notional, which in a swap is never at risk, nor the portfolio's value, which can be negative. It varies with the date, and collateral held against it reduces it; the definition below leaves collateral out until equation (9.2) subtracts it. If a counterparty defaults at time t, the loss depends on what the portfolio is worth at that moment. If the portfolio has *negative* value (the dealer owes money), there is no loss; the obligation survives the counterparty's default and must still be paid. If the portfolio has *positive* value, the dealer joins the queue of unsecured creditors and recovers a fraction.

The exposure is therefore

$$E(t) = \max\left(V(t),\, 0\right) \tag{9.1}$$

where $V(t)$ is the value at time t of the trades held with that counterparty, taken together and signed from the dealer's point of view, so that $V(t) > 0$ means the counterparty owes the dealer. Which trades may be taken together is a legal question, taken up below. Equation (9.1) is the payoff of a call option on that value struck at zero: *counterparty risk is an option*, and everything that makes options hard (volatility, correlation, the need to model a distribution rather than a point) applies to it.

Three summaries of the exposure distribution are used:

- **Expected exposure** (EE), the average over scenarios at each future date: $\mathrm{EE}(t)$ is the average of $\max(V(t), 0)$.
- **Expected positive exposure** (EPE), the average of $\mathrm{EE}(t)$ over the life of the trade, a single number summarizing the profile. Banks using the internal model method capitalize a variant, effective EPE (Section 27.5).
- **Potential future exposure** (PFE), a high quantile of the exposure distribution, used for credit limits rather than valuation.

How exposure is computed

Obtaining $\mathrm{EE}(t)$ means obtaining the distribution of the portfolio's value at every future date, by a Monte Carlo simulation in three stages.

1. *Evolve the risk factors.* Simulate paths of the underlying risk factors (interest rate curves, FX rates, credit spreads, commodity prices) from today to T on a chosen time grid.
2. *Revalue the portfolio.* On every path, at every grid date, revalue every trade in the netting set. This is the expensive step: N paths $\times$ m dates $\times$ K trades valuations.
3. *Aggregate.* On each path and date, net across trades, subtract collateral, take the positive part, and average across paths.

   Formally, for path j and date t_i,

$$E_j(t_i) = \max\left(\sum_{k=1}^{K} V_k^{(j)}(t_i) - C^{(j)}(t_i),\; 0\right), \qquad \mathrm{EE}(t_i) = \frac{1}{N}\sum_{j=1}^{N} E_j(t_i) \tag{9.2}$$

where $C^{(j)}(t_i)$ is the collateral held on that path and date.

The maximum in (9.2) is taken *after* netting across trades and subtracting collateral. Taking it trade by trade and summing overstates exposure, by convexity (the netting inequality (9.4) below), in some portfolios by a large multiple.

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Computing cost is what limits the calculation, which is among the largest a dealer runs: 5,000 paths and 100 dates on a netting set of 2,000 trades is a billion trade valuations for one counterparty, repeated across thousands of counterparties nightly. Two techniques make it feasible. **American Monte Carlo** (the least-squares regression of Section 20.11, borrowed for its regression step rather than for the early-exercise decision) replaces exact revaluation with a fitted function of the state variables. **Adjoint algorithmic differentiation** (AAD, Section 20.10) produces every sensitivity in a small multiple of one valuation's cost, and the desk of Section 9.10 hedges the charge's sensitivity to every market input, not only its level.

The shape of an exposure profile

For an interest rate swap the exposure profile has a characteristic hump. Two forces work against each other. *Diffusion*: the longer rates have had to move, the wider the range of possible values, which pushes exposure up roughly with $\sqrt{t}$. *Amortization*: as payment dates pass, the swap's sensitivity to rates falls, which pulls exposure down roughly linearly. The product peaks at about one third of the way through the trade.

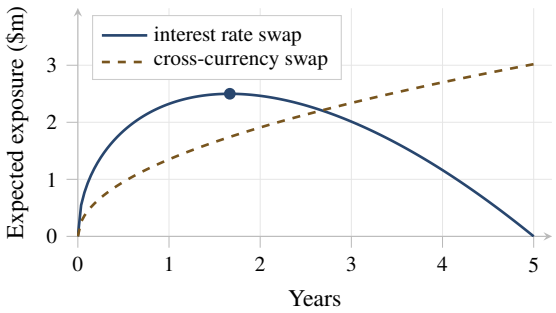


Figure 9.1: Expected exposure for a five-year interest rate swap and for a cross-currency swap (the latter schematic).

Figure 9.1 contrasts this with a cross-currency swap. There the notional is exchanged at maturity (Chapter 7), so the largest cash flow is the last one and nothing amortizes: diffusion operates unopposed and exposure rises to maturity. A cross-currency swap therefore carries more counterparty risk than an interest rate swap of the same notional and tenor, and carries it later, when the counterparty has had longest to default. The curve drawn is schematic and understates the difference (on Example 9.2's inputs the two curves give CVAs only about 9% apart), because the final exchange of notional is exposed to the whole move in the exchange rate, which at ordinary currency volatilities puts a real profile well above the one drawn. Between dealers the effect is usually contained by a mark-to-market resetting cross-currency swap, which adjusts one notional to the current exchange rate at each reset and so returns the currency move to the counterparty period by period (Section 7.8; Chapter 33 prices it); the corporate swaps that carry most cross-currency CVA rarely reset.

Other products have other shapes. A purchased option has exposure equal to its full value throughout, since that value can never be negative, while a sold option has none once the premium is received, so two trades that look symmetric carry radically asymmetric counterparty risk. An FX forward rises with $\sqrt{t}$ to maturity like the cross-currency swap, because nothing amortizes. A repo or fully collateralized financing is exposed only to the extent that the collateral can lose more than the haircut over the margin period of risk, which is why Chapter 8's haircuts belong to credit as well as to liquidity. In general, exposure grows with the uncertainty remaining and shrinks with the obligations discharged.

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Expected exposure and potential future exposure

Expected exposure and potential future exposure serve different purposes, and confusing them produces numbers a factor of four apart (Example 9.1). Suppose that at the peak of the profile the portfolio value is normally distributed about zero with standard deviation σ. Then the expected exposure has a closed form,

$$\text{EE} = \int_0^\infty v \, \frac{1}{\sigma\sqrt{2\pi}} \, e^{-\frac{v^2}{2\sigma^2}} \, dv = \frac{\sigma}{\sqrt{2\pi}} = 0.3989\,\sigma \tag{9.3}$$

while the potential future exposure at the 95% level is the corresponding quantile of the positive part, $1.645\,\sigma$.

Example 9.1 Expected against potential future exposure

The swap of Figure 9.1 peaks at an expected exposure of \$2.5 million. From (9.3) the implied standard deviation of the portfolio's value at that date is

$$\sigma = \frac{2.5}{0.3989} = \$6.27 \text{ million}$$

so the 95% potential future exposure is

$$\text{PFE}_{95} = 1.645 \times 6.27 = \$10.3 \text{ million}$$

roughly four times the expected exposure.

Both numbers describe the same trade, and both are used, for different things. *Pricing* uses expected exposure, because the hedge of Section 9.2 is sized to the average and its cost is what the client is charged. *Credit limits* use potential future exposure, because a limit is meant to cap a bad case rather than an average one. In practice the two are usually simulated differently as well: exposure for pricing from risk factors calibrated to market prices such as implied volatilities, exposure for limits from dynamics estimated on history.

A credit officer refusing a trade on a \$10.3 million PFE and a trader pricing it off a \$2.5 million EE are not disagreeing; they are answering different questions.

Netting matters, and it matters at the right level. Exposure is computed on the netting set (the group of trades covered by one master agreement), not trade by trade:

$$E(t) = \max\left(\sum_i V_i(t),\, 0\right) \;\le\; \sum_i \max\left(V_i(t),\, 0\right) \tag{9.4}$$

The inequality is strict whenever some trades have positive values and others negative ones, so a new trade's contribution to exposure can be *negative*: a trade that offsets the existing portfolio reduces the netting set's exposure and with it the CVA charge. In the limiting case, a client that already pays fixed on a swap and now asks to receive fixed on identical terms closes the netting set's exposure to zero, and the new trade's incremental CVA is minus the whole CVA of the existing one. A dealer that quotes a better price for a trade that hedges its existing exposure to a client than for one that adds to it is not giving a discount; it is pricing correctly.

9.2 CVA

The **credit valuation adjustment** is the market value of the counterparty's option to default: the amount by which a trade is worth less than the same cash flows promised by a counterparty that cannot fail. Everything in this book is priced the same way: the value of a position is what it costs to hedge it, a principle Chapter 13 makes exact. Here it applies directly, because the hedge for this option already trades. A credit default swap on the counterparty (introduced in Chapter 1, named in Chapter 7, and analyzed in Chapter 25) pays its buyer the loss on a stated

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amount if that counterparty defaults, in exchange for a running premium, so the premium rate, the **CDS spread**, is the market's price for insuring one dollar owed by that counterparty for one year. CVA is the cost of insuring the exposure. No probability is estimated; the spread is a price, and it is used as one. The formula is worth building rather than quoting, because everything questionable about CVA enters as an assumption along the way.

What a default costs. Suppose the counterparty defaults at some time τ before the final maturity T. Write $V(t)$ for the portfolio's value assuming no default, and R for the **recovery rate**, the fraction of its claim that a creditor eventually collects from the estate, so that a default costs it $1 - R$, the fraction the credit chapters call the loss given default. Recovery is treated here as a known constant, which it is not: it varies with seniority and with the state of the economy, and Chapter 25 describes how the market fixes it by auction. As Section 9.1 showed, the loss is one-sided, which is what makes it an option:

$$L = (1 - R)\max\left(V(\tau),\, 0\right) \quad \text{if } \tau \le T, \qquad \text{and zero otherwise.} \tag{9.5}$$

What insuring it costs. A CDS with notional N pays $(1 - R)N$ on default, the same $1 - R$, because the protection seller makes good the loss on a bond of that face amount. A dealer that knew the exposure path in advance could buy protection on an amount equal to $E(t)$ for each future date, and the payout would match the loss in (9.5) exactly. The premium is the spread s times the amount covered, per year, paid only while the counterparty is still alive. The cost of the hedge is therefore a stream of premiums, $s \times E(t)$ a year, running until maturity or default, whichever comes first, and CVA is the present value of that stream. Two assumptions turn this into something a desk can compute.

Assume the amount to insure is the average exposure. The exposure is a distribution at each date, not a known path, and the dealer insures its average, the expected exposure $\mathrm{EE}(t)$. That is the right amount only if the size of the exposure when the counterparty fails is unrelated to the failure itself; where the exposure is largest in the states in which the counterparty is likeliest to fail, the average understates what needs cover (Section 9.9).

Assume discounting separates from exposure. This is exact under fixed rates and otherwise a good approximation, the link between the discount factor and the exposure being second order.

Discretizing on a grid, $0 = t_0 < t_1 < \cdots < t_n = T$, and writing $c(t_{i-1}, t_i)$ for the cost of covering one dollar of exposure over the interval (the premium for that interval, scaled by the fraction of scenarios in which the counterparty is still alive to be insured), and $\mathrm{DF}(t_i)$ for the discount factor to t_i:

$$\boxed{\mathrm{CVA} = \sum_{i=1}^{n} \mathrm{EE}(t_i)\; c(t_{i-1}, t_i)\; \mathrm{DF}(t_i)} \tag{9.6}$$

The spread is the *counterparty's*: CVA prices the risk that it fails, and Section 9.3 turns the same calculation on the dealer's own. It comes from market spreads (the counterparty's own CDS or bonds where they trade, otherwise a proxy curve for its rating, sector, and region) rather than from historical default rates.

Chapter 24 arrives at the same cost from the other side. A spread of s with a loss given default of $1 - R$ means the market is charging as though the counterparty fails at a rate of about $\lambda = s/(1 - R)$ a year, a relation Chapter 24 calls the credit triangle; with that rate, the cost of cover for an interval is the loss on default times the chance of a default falling in it,

$$c(t_{i-1}, t_i) = (1 - R)\, q(t_{i-1}, t_i), \qquad q(t_{i-1}, t_i) = e^{-\lambda t_{i-1}} - e^{-\lambda t_i}.$$

The two readings must agree, because a CDS is priced so that its premiums and its payouts balance, and Example 9.2 checks that they do.

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Example 9.2 Computing CVA on a swap

A dealer has an uncollateralized five-year swap with a corporate client on a notional of $100 million. The expected exposure profile is the interest rate swap curve of Figure 9.1, and equation (9.3) produces it. By date t the swap's value has moved by roughly the change in the swap rate times the notional L times the annuity of the payments still to come, about $T - t$ years of them; the rate change has a standard deviation of $\sigma_r \sqrt{t}$, so the value is roughly normal about zero with standard deviation $L\,\sigma_r \sqrt{t}\,(T - t)$, and

$$\mathrm{EE}(t) = 0.3989 \times L\,\sigma_r \sqrt{t}\,(T - t)$$

With L = $100 million, T = 5 years and a rate volatility σ_r of 1.456% a year, the profile rises with $\sqrt{t}$ as the rate diffuses and falls as the swap amortizes, and at mid-year points it gives the expected exposure column of the table below. Section 20.6 confirms it by simulation, at the same volatility. The client's CDS trades at 120 basis points a year; recovery is assumed to be 40%, so the loss given default is 60%, and the credit triangle puts the market's default rate for this issuer at $0.012/0.60 = 2\%$ a year. Discounting is at 4% continuously compounded.

Equation (9.6) is evaluated on an annual grid, using mid-year exposures and discount factors. The cost of cover in each year is the 120 basis point premium scaled by the fraction of scenarios in which the client is still alive to be insured, $e^{-0.02\times0.5} = 0.990$ in the first year and falling by 2% a year after that:

Year	EE ($m)	Cost of cover	Discount factor	Contribution ($m)
1	1.848	1.188%	0.9802	0.02152
2	2.490	1.165%	0.9418	0.02731
3	2.296	1.141%	0.9048	0.02371
4	1.630	1.119%	0.8694	0.01585
5	0.616	1.097%	0.8353	0.00564
			CVA	0.09404

$$\mathrm{CVA} = \$94{,}000$$

That is 9.4 basis points of notional, charged up front. It is not a reserve the dealer sets aside from prudence; it is the market price of an option the dealer has written (the client's option to walk away), and if the dealer does not charge it, the trade is mispriced from inception.

The cost-of-cover column is also what Chapter 24's reading gives. In the first year the chance of a default is $1 - e^{-0.02} = 1.980\%$, and $0.60 \times 1.980\% = 1.188\%$, the figure in the table; the same holds in every row to the third decimal. Insuring the exposure and bearing the expected loss on it cost the same, as they must when the insurance is fairly priced.

Years 2 and 3 dominate, because that is where the exposure hump sits. A trade whose exposure peaked later, like the cross-currency swap of Figure 9.1, would carry a larger CVA for the same notional and the same counterparty: about $102,000 with that curve as drawn, some 9% more, and more again at realistic currency volatility.

Why the spread, and not the record. The cost of cover in the table is about 1.2% a year. Borrowers that trade at that kind of spread have, historically, cost their creditors much less, by a factor of several for an issuer of this quality and far more for the safest credits (Section 24.9). Concluding that the CDS market overprices, and charging the client the historical figure, would be a mistake. The spread is not a forecast of how often the counterparty will fail; it is what a seller of protection demands for a risk that pays off in exactly the bad times, when the seller least wants to pay, and so it includes compensation for that timing, for the seller's own funding and capital, and for the possibility that this counterparty turns out worse than its history. CVA uses the spread because CVA is what the hedge costs, not what the loss is expected to be. A dealer that charged historical CVA would undercharge by that same several-fold factor, and

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would find every hedge it bought costing more than the revenue meant to pay for it. Chapter 28 shows that the same is true of every price in this book: derivatives are priced by what they cost to replicate, and the probabilities a hedging cost implies are not the probabilities a statistician would estimate.

9.3 DVA

If the counterparty's option to default has value, so does the dealer's. The **debit valuation adjustment** (DVA) is the mirror image of CVA: the value to the dealer of its own option to default, computed on the *negative* exposure (the states in which the dealer owes money and would not pay it) at the dealer's own CDS spread, exactly as the counterparty would compute CVA on the dealer.

DVA has an impeccable derivation and an absurd implication. It is required for symmetry: if the client charges the dealer CVA for the dealer's credit risk, and the dealer charges the client CVA for the client's, then each party's CVA is the other's DVA, and the two agree on the trade's value. The implication is that a deterioration in the dealer's own credit is a gain. As the dealer's spread widens, its own option to default becomes more valuable, DVA rises, and the dealer books a profit.

Case Study 9.1 Profiting from your own decline

In 2011, as bank credit spreads widened sharply in the euro area crisis, several large banks on both sides of the Atlantic reported substantial DVA gains. The accounting was correct and the logic unimpeachable: their liabilities, including their derivative liabilities, had become less likely to be paid in full and were therefore worth less, and a fall in the value of a liability is a gain.

The presentational problem was obvious (institutions reporting profits derived from the market's declining confidence in them), and the prudential problem was worse. Capital exists to absorb losses, and capital that increases as the firm's creditworthiness falls does the opposite: a firm approaching failure would report its largest DVA gains just as its capital was most needed.

Regulators resolved this by filtering DVA out of regulatory capital: the gain may be recognized in accounting profit but is deducted from common equity. The effect is a permanent gap between the accounting value of a derivatives book and its regulatory value.

Much of the 2011 gain was not on derivatives at all but on the banks' own bonds carried at fair value, and the accounting standards have since moved that part out of profit and into other comprehensive income (IFRS 9 and, in the United States, Accounting Standards Update (ASU) 2016-01, both required from 2018). The DVA on derivative liabilities still runs through profit.

Most dealers treat DVA as real for valuation and hard to use for management. It cannot be hedged directly, since that would mean selling protection on oneself, which no one will buy in the size required; banks have hedged it approximately by selling protection on peer banks whose spreads move with their own, taking on those banks' credit. And it is realized, if at all, mostly by defaulting, or in part through the funding benefit of Section 9.4, where the two adjustments overlap. It therefore appears in the accounts and carries little weight in pricing.

Two refinements matter for a precise bilateral calculation. Each party's charge should be conditioned on the other surviving (the first-to-default effect), which makes bilateral CVA and DVA each slightly smaller than their one-sided versions. And the amount claimed on default depends on the close-out convention: if the surviving party may claim the cost of replacing

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the trade with a new counterparty, that replacement value includes the surviving party's own DVA, and the loss on default is smaller than a risk-free close-out implies. The 2002 ISDA Master Agreement frames its close-out amount as a replacement cost, which points toward the replacement reading, though how far in practice is not settled.

9.4 FVA

The **funding valuation adjustment** is the cost of funding an uncollateralized derivative position, and it is the most contested item in this chapter. Section 8.1 gave the mechanism: the dealer hedges the uncollateralized client trade with a margined one, so when the client trade moves in the dealer's favor the hedge calls variation margin in cash that nothing from the client offsets, and the difference has to be borrowed. If the trade is against the dealer, the dealer keeps cash that a CSA would have required it to hand over, while the hedge returns margin. Either way the trade leaves a funding position (a cost in the first case, a benefit in the second), carried at the dealer's own cost of funds, which is above the risk-free rate by its funding spread. Under the one-way CSA of Section 8.1, which sovereigns, supranationals and central banks typically insist on, the dealer must pass to the client the margin its hedge returns, so the second case disappears and FVA is all cost.

Example 9.3 Computing FVA

Take the swap of Example 9.2 and suppose the dealer's funding spread over the risk-free rate is 60 basis points. The discounted expected exposure uses the exposure and discount columns of Example 9.2 with the cost of cover left out, each year weighted by its length $\Delta t_i = 1$:

$$\sum_i \mathrm{EE}(t_i)\,\mathrm{DF}(t_i)\,\Delta t_i = 1.848 \times 0.9802 + 2.490 \times 0.9418 + 2.296 \times 0.9048$$
$$+ 1.630 \times 0.8694 + 0.616 \times 0.8353 = \$8.17 \text{ million-years}$$

The funding cost is

$$\mathrm{FVA} = 0.0060 \times \$8.17\text{m} = \$49{,}000$$

So on this trade FVA is about half of CVA. This is the cost side alone, the funding of positive exposure; the benefit on negative exposure is taken up below. The ratio is not general. It is roughly the funding spread against the counterparty's credit spread, here 60 basis points against 120, and it comes out a little above one-half because the cost of cover decays with survival while the funding cost, as computed here, does not. But the example makes the point that funding is the same order of magnitude as credit, not a rounding item.

Case Study 9.2 The FVA debate

In 2012 Hull and White argued, in a widely read article, that FVA should not exist.

Their argument applies a principle this book has used throughout: the value of an asset is determined by its risk, not by how the buyer chose to finance it. A firm with a high funding cost does not get to charge more for the same cash flows; that the cheapest-funded firm would win every trade is a statement about competition, not about value. Charging FVA, on this view, confuses the dealer's funding problem with the trade's worth, and amounts to passing the dealer's own credit risk to the client twice, since the client is already compensated through DVA.

The practitioner response is that this is correct in a frictionless world and irrelevant in this one. A dealer that ignores FVA and prices at theoretical value will win exactly the trades that are expensive for it to fund, and will discover the cost in its funding line rather than in its trading book. Whatever the theory says, the cash outflow is real.

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The debate was settled, in the way such debates usually are, by accounting. Firms began recognizing FVA in their published results (most visibly when JPMorgan took a $1.5 billion FVA charge in the fourth quarter of 2013), and once one major dealer had recognized it, the others followed rather than explain the discrepancy.

The theoretical question was answered later, and the answer gives each side part of what it claimed. Andersen, Duffie, and Song (2019) showed that funding an uncollateralized position transfers value from shareholders to the bank's creditors, so FVA is a real cost to the shareholders who decide whether to trade, and belongs in the price the desk quotes; but it is not a property of the cash flows, and the trade's value to all of the bank's stakeholders together does not contain it. It is worth understanding both positions, because the same argument recurs whenever a cost that is real to a firm is not a property of the instrument it trades.

FVA and DVA overlap. A genuine technical problem underlies the dispute. The dealer's funding spread reflects its own credit risk, which DVA also prices, so charging both counts the same thing twice, at least in part. Practice varies: some firms compute FVA using only the liquidity component of their funding spread; others restrict FVA to a funding cost adjustment on positive exposure and set the corresponding funding benefit adjustment on negative exposure against DVA. No treatment is universally agreed, and two dealers can quote materially different prices on identical trades for this reason alone.

9.5 MVA

The **margin valuation adjustment** is the cost of funding initial margin over the life of a trade. It exists because of the rules Chapter 8 described: initial margin is posted gross, segregated, and cannot be reused, so it must be funded from inception to maturity.

Example 9.4 Computing MVA

Suppose the swap of Example 9.2 is instead cleared. The clearing house sizes initial margin to cover the swap's value change over a five-day close-out at a high confidence level, 99% or more; at the rate volatility behind Figure 9.1 the five-day standard deviation of the value is about $1 million, so the margin starts at about $2.5 million and declines roughly linearly to zero as the trade amortizes, an average of $1.25 million. MVA is charged on the margin the trade adds to the dealer's cleared portfolio, and this dealer's cleared book is assumed not to offset the trade, so the increment is the whole stand-alone amount; Example 8.5's dealer, whose cleared leg nets against its existing book, would pay far less. The dealer funds the margin at the same 60 basis point spread.

A charge accruing at a constant rate over the trade's life is worth that rate times the trade's discounted life, which at the 4% used throughout this chapter is

$$\int_0^5 e^{-0.04t}\,dt = \frac{1 - e^{-0.2}}{0.04} = 4.53 \text{ years}$$

Using that discount-weighted duration,

$$\text{MVA} = 0.0060 \times \$1.25\text{m} \times 4.53 = \$34{,}000$$

Compare this with the figures already computed: CVA of $94,000 and FVA of $49,000 (Examples 9.2 and 9.3). On this trade the cost of funding initial margin is about a third of the credit charge it replaces.

The proportion is not fixed. MVA scales with the margin model's output and with the funding spread, and both rise in exactly the conditions that raise everything else: a stressed lookback period in the margin model, a ten-day close-out for an uncleared trade instead of five, a longer maturity, or a dealer whose own funding has become expensive. Practice Question 8 doubles the margin and raises the spread by half, and MVA overtakes CVA. Clearing and the uncleared margin rules

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> replaced part of the credit risk they removed with a funding requirement, and a firm that dropped its CVA desk after mandatory clearing, on the grounds that counterparty risk had gone, would have been charging nothing for a cost that had changed form rather than disappeared.

MVA is also awkward to compute, because it requires forecasting a *margin model's output* on future dates and scenarios: a clearing house's own initial margin model for a cleared trade, SIMM for an uncleared one under the rules of Section 8.4. Under the uncleared rules the cost is also marginal in a way the cleared one is not: two firms exchange no initial margin until the requirement between them passes the €50 million threshold, so a trade that fits under it carries no MVA, and once the threshold is used up every further trade carries its full margin.

On the same grid as (9.6),

$$\text{MVA} = \sum_{i=1}^{n} s_F(t_i)\, \overline{\text{IM}}(t_i)\, \text{DF}(t_i)\, \Delta t_i \tag{9.7}$$

where $s_F(t)$ is the dealer's funding spread over the rate paid on posted margin (60 basis points in Example 9.4), $\overline{\text{IM}}(t_i)$ is the initial margin outstanding at t_i, averaged across scenarios, and $\Delta t_i = t_i - t_{i-1}$. Like the other adjustments, this is written as a positive cost; Section 9.8 carries the signs.

The difficulty in (9.7) is entirely in $\overline{\text{IM}}(t_i)$: computing it exactly means running the margin model *inside* the Monte Carlo, evaluating sensitivities on every path and date and aggregating them through the margin formulas, which multiplies an already expensive calculation by the cost of a full sensitivity run. Three approximations are used.

A deterministic profile scales today's initial margin by the remaining risk (for a swap, roughly by remaining duration). That is what Example 9.4 does with its linear decline, and it is adequate while the portfolio's composition is stable.

Regression fits IM as a function of the simulated state variables on a subset of paths and evaluates the fitted function everywhere: the regression Section 9.1 borrowed from American Monte Carlo for exposure, turned on the margin model rather than on value.

Sensitivity scaling applies the model's risk weights to the sensitivities simulated on each path, but replaces the step that combines them through the model's correlation formulas, which depends on the whole portfolio on that path, with a simpler fixed rule.

All three approximate a quantity that is itself an approximation, a standardized margin model being no more than a proxy for a risk measure.

9.6 COLVA AND THE TERMS OF THE CSA

Chapter 8 showed that a CSA is rarely the idealized arrangement in which collateral exactly tracks the mark-to-market in cash, daily, in one currency. Real CSAs have thresholds, minimum transfer amounts, non-cash eligible collateral, and currency choice, and each departure has a value. Strictly, the **collateral valuation adjustment** (ColVA) is the value of the terms on which collateral is remunerated and chosen: a collateral rate (or, for non-cash collateral, a return) other than the overnight rate the curve is built on, a floor on that rate, and the right to choose what to post, such as the cheapest of several currencies, an option over the discount curve that Example 8.6 values. Thresholds, minimum transfer amounts, and the margin period of risk (the gap between the last collateral movement and close-out) act differently. They leave exposure uncovered, a $5 million threshold being a $5 million unsecured credit line, so their cost is CVA and FVA computed on the exposure that remains; adding it again under ColVA would count it twice. Desks discuss the whole set together, as the terms of the CSA, and so does this section. Benchmark reform moved most CSAs' cash rate from EFFR or EONIA to SOFR or €STR, and since 2016 the uncleared margin rules have required daily variation margin between the firms

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they cover, so the standard CSA has drifted toward the idealized one and the residual terms matter most on legacy agreements.

Thresholds, minimum transfer amounts, and the margin period of risk enter the simulation through one term, the $C^{(j)}(t_i)$ of (9.2), and the modeling has a single essential feature: collateral reflects the portfolio's value at an earlier date. Writing δ for the margin period of risk, H for the threshold and M for the minimum transfer amount, and setting to zero the independent amount that Chapter 8's credit support amount also includes,

$$C(t) = \max\left(V(t-\delta) - H,\ 0\right) \quad \text{subject to changes of at least } M \tag{9.8}$$

Equation (9.8) describes collateral the dealer *holds*. Under a two-way CSA the dealer also posts when V is negative, symmetrically, and posted collateral matters in two ways the equation omits: it is an exposure of its own if the counterparty does not segregate it, and it must be funded, which is where FVA enters a collateralized trade.

The consequence is that a perfect CSA does not give zero exposure. Even at $H = M = 0$, equation (9.8) leaves $C(t) = \max(V(t-\delta), 0)$: the collateral held reflects the value one margin period ago, so the collateralized exposure is at most the portfolio's move over that window, and exactly that whenever the dealer was holding collateral,

$$E(t) = \max\left(V(t) - V(t-\delta),\ 0\right)$$

At Chapter 8's ten days bilateral and five days cleared, this residual is why a fully collateralized trade still carries a small CVA rather than none. Initial margin is sized to cover exactly this move, which is what it is for, and why it costs what Section 9.5 says it does.

Example 9.5 The cost of a threshold

The swap of Example 9.2 is with a corporate client outside the uncleared margin rules, which would otherwise require a zero threshold (Chapter 8). The client insists on a \$5 million threshold in its CSA, below which no collateral moves. The dealer agrees, believing the CSA to be otherwise standard.

For exposures up to \$5 million the trade is effectively uncollateralized. In most scenarios the exposure stays below that level, so on the normal model of Section 9.1 the expected exposure capped at \$5 million averages about \$1.4 million over the mid-year dates, against \$1.78 million uncapped. Insuring that amount at the client's 120 basis point spread for the trade's discounted life carries

$$\text{CVA on the threshold} \approx 0.012 \times \$1.4\text{m} \times 4.53 = \$76{,}000$$

plus funding on the same amount at 60 basis points,

$$0.0060 \times \$1.4\text{m} \times 4.53 = \$38{,}000$$

A concession that sounded operational ("we would rather not exchange small amounts") costs about \$114,000 on a single trade, most of the way to the uncollateralized CVA and FVA. (Weighting the premiums by survival, as Example 9.2 does, would take a few thousand dollars off; at this precision it does not matter.) Thresholds are priced, and clients are frequently surprised by how much.

9.7 KVA

The **capital valuation adjustment** (KVA) is the cost of the regulatory capital a trade requires over its life: the equity a bank must hold against the position because the rules say so, rather than because it chose to. How much capital a trade requires is Chapter 27's calculation (counterparty exposure under the standardized approach for counterparty credit risk (SA-CCR) in Section 27.3, the charge against CVA's own volatility in Section 27.6), and Example 9.6 takes the result as given.

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Capital is not free. Shareholders require a return above the risk-free rate (a hurdle of perhaps 8 to 12% over it), and a trade that consumes capital for ten years consumes that return for ten years. If the trade does not earn it, the shareholders are worse off even though the trading book shows a profit.

Example 9.6 Computing KVA, and its size

Example 27.3 computes the counterparty credit capital for the uncollateralized swap of Example 9.2 at $0.33 million at inception, against a corporate at a 100% risk weight. Counterparty capital rises in the first years as replacement cost builds, then falls as the swap runs off. The example takes the case in which the client, a nonfinancial corporate, is exempt from the CVA capital charge, as it is under the EU rules (Section 27.7), so the capital is counterparty credit capital alone; suppose it averages $0.4 million over the swap's life, and that the firm's hurdle rate over the risk-free rate is 10%. Then

$$\text{KVA} = 0.10 \times \$0.4\text{m} \times 4.53 = \$181{,}000$$

This is about twice the CVA and nearly four times the FVA. Where the CVA capital charge applies, as it does to the same trade in the United States and Canada (Section 27.7), it adds roughly as much capital again and KVA is materially higher. On long-dated uncollateralized trades KVA is often the largest adjustment of all, and it is the reason such trades became expensive and less common after the post-crisis capital rules took effect. Practitioners compute it on a projected capital profile, trade by trade within the portfolio, and include the leverage ratio where it binds, since a bank must hold capital against whichever rule is the stricter.

CVA the price and CVA the capital charge

The same three letters denote two different things, a persistent source of confusion.

CVA as used in Section 9.2 is a *price*: the market value of the counterparty's default option, deducted from the value of the trade.

The **CVA capital charge** is a *regulatory requirement*: capital held against the risk that CVA itself *moves*. Its rationale is empirical: in the financial crisis, by the Basel Committee's count, roughly two-thirds of losses on counterparty exposures came not from counterparties actually defaulting but from the mark-to-market deterioration of CVA as credit spreads widened. Firms held capital against default and were hit by volatility.

The charge is computed under the Basel framework of Section 1.12, either by the basic approach (BA-CVA), a formula on exposures and supervisory risk weights, or, with supervisory approval, by the standardized approach (SA-CVA), which works from CVA's sensitivities to credit spreads and to the market factors driving the exposure. The Basel standard took effect in 2023, in the European Union in 2025 and in the United Kingdom in 2027; the US agencies re-proposed theirs in March 2026, and they were not final at the time of writing. Section 27.6 sets out the approaches and Section 27.7 the exemptions. The charge feeds into KVA, since it is part of the capital the trade consumes.

The distinction matters because the two respond to different actions. Single-name CDS protection on a counterparty hedges the price and reduces the charge (under BA-CVA by at most three-quarters), and novating the trade to a clearing house eliminates both. But for most counterparties no single-name CDS trades (Section 9.10), and the index or proxy protection a desk can actually buy hedges the *price* while earning only partial relief from the *capital charge*. Tightening a CSA lowers both by different amounts: the price by what the firm's own exposure model says, the charge by what the standardized formulas allow.

KVA is the most conceptually troubled of the adjustments. Capital requirements are set by regulators and change; the hurdle rate is a management choice, not a market price; and including a firm's cost of equity in the valuation of an instrument sits uneasily with the idea

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that value is a property of the instrument. Some firms treat KVA as a pricing hurdle (a floor below which they will not trade) rather than a valuation adjustment carried in the accounts, which is arguably the more honest treatment.

9.8 PUTTING THEM TOGETHER

Table 9.1 collects the adjustments for the trade used throughout this chapter, under two collateral arrangements. Two entries rest on inputs the examples did not state. DVA is computed at the dealer's own spread, taken as 49 basis points, on the mirror image of the client's exposure profile: it assumes the dealer's negative exposure has the same profile as its positive one, which holds for an at-market swap under the symmetric normal model of Example 9.2. And the cleared trade's capital is taken as averaging $28,000, set by the leverage ratio rather than by the risk-based rule: Example 27.3's trade exposure to the clearing house needs only about $1,950 at the 2% risk weight, but at a 3% leverage requirement it needs about $28,000 (Section 27.10). The totals are net of DVA.

Table 9.1: XVA on a five-year $100m swap under two collateral regimes, net of DVA.

Adjustment	Uncollateralized	Cleared	Driver
CVA	−94,000	≈ 0	counterparty default
DVA	+39,000	≈ 0	own default
FVA	−49,000	≈ 0	funding the exposure
MVA	—	−34,000	funding initial margin
KVA	−181,000	−13,000	regulatory capital
Total	**−285,000**	**−47,000**	

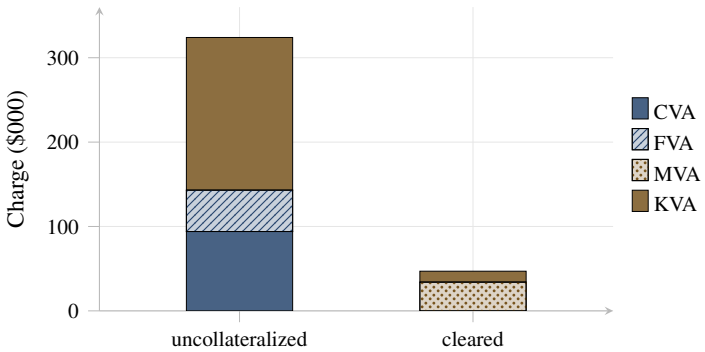


Figure 9.2: Composition of the charge in Table 9.1, gross of DVA.

Figure 9.2 plots the composition, and shows what the totals conceal: the two columns are not the same charge in different sizes but largely different charges. Three observations follow.

The adjustments are large. On the uncollateralized trade they total 28.5 basis points of notional net of DVA. DVA carries little weight in pricing (Section 9.3), so the charge a client is actually quoted is nearer the $324,000 before it: 32.4 basis points of notional, or about 7 basis points a year on the swap rate, against an interdealer bid–ask spread on a five-year swap of a fraction of a basis point. The XVA is not a correction to the price; on this trade it substantially *is* the price.

Clearing changes the composition of the charge rather than eliminating it. Credit and funding charges collapse toward zero, and are replaced by the cost of funding initial margin. The total falls (clearing is genuinely cheaper, as Chapter 8 explained it was designed to be),

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but not to zero. For this client the cleared column is a comparison rather than a choice: a nonfinancial corporate hedging its own business is usually exempt from mandatory clearing and from the uncleared margin rules (the end-user exception in the United States, the clearing threshold in the European Union), which is why uncollateralized corporate swaps still exist and why the left-hand column is the one a corporate treasurer is quoted.

Capital dominates. On the uncollateralized trade KVA is larger than everything else combined. That explains market structure changes that otherwise look puzzling: why dealers pushed clients into clearing, why long-dated uncollateralized corporate trades became expensive, and why balance-sheet-light business models proliferated.

9.9 WRONG-WAY RISK

Wrong-way risk arises when exposure increases in the same states in which the counterparty is more likely to default. The textbook example is buying credit protection on a sovereign from a bank in that country: the protection is worth most when the sovereign is in trouble, which is exactly when the bank is least able to pay. Chapter 25 shows this was not hypothetical.

Less dramatic cases are common. A dealer that has bought protection against falling oil prices from an oil producer has exposure that rises when oil prices fall, which is when the producer's credit is worst. A dealer receiving fixed from a highly levered borrower has exposure that rises as rates fall, which may correlate with the conditions that stress the borrower.

Right-way risk is the reverse and reduces CVA: a dealer that has bought protection against *rising* oil prices from the producer has exposure rising when oil rises, which is when the producer is strongest.

The insurance argument shows exactly where the first assumption of Section 9.2 did its work. What the dealer needs cover for is the exposure *at the moment the counterparty fails*, the expected exposure *given that* the default falls in each interval, and the standard calculation replaces it with the unconditional average. Under wrong-way risk the exposure in the scenarios where the counterparty actually defaults is systematically larger than the average across all scenarios, so the amount insured is too small, and (9.6) is a lower bound rather than an estimate.

Two modeling approaches are standard, and regulation avoids the question:

- *Intensity models* let the counterparty's default rate rise and fall with the driver of the exposure.
- *Copula approaches* join the default time and the exposure while leaving each one's own distribution as it was, the device Section 26.6 turns on portfolio credit.
- *Regulation* sidesteps both, covering general wrong-way risk with the multiplier of 1.4 (Section 27.5) and specific wrong-way risk with a rule that treats a legally connected reference entity as already defaulted (Section 27.9).

Example 27.4 puts a size on the effect: a dependence strong enough to double the exposure expected at default doubles the CVA. Where the dependence is definitional rather than statistical, no correlation parameter is adequate. Protection bought on a sovereign from a bank domiciled in it is not an exposure with a correlation adjustment; the exposure and the default are the *same event*, and the trade should be assessed as a jointly contingent claim.

9.10 THE XVA DESK

The organizational response to all this is a central **XVA desk**. When a trader books a client trade, the XVA desk charges the relevant adjustments and takes the corresponding risk onto its own book. The originating desk is left with a clean, hedgeable market risk position; the XVA desk holds the credit, funding, and capital exposures across the whole firm. The centralization exists because these risks are *portfolio* properties, not trade properties: exposure is a netting-set

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quantity (9.4), so a new trade's CVA depends on everything already facing that counterparty, and only a desk that sees the whole book can price the increment.

Hedging. The XVA desk hedges CVA by doing what Section 9.2 priced: buying credit protection on its counterparties, and hedging the sensitivity of the exposure itself to rates and FX. Because the exposure is an option, CVA has a delta and a vega: sensitivities to the underlying market variable and to its volatility. Hedging CVA is harder than it sounds: liquid CDS exists for a few hundred issuers, while a dealer faces thousands of counterparties, most with no traded credit, so much CVA is hedged by proxy, with index CDS and correlation assumptions, leaving basis risk whenever a specific issuer diverges from its proxy.

Other instruments fill parts of the gap. A **contingent CDS** pays the loss on a notional that is itself defined by a referenced derivative's value at default, so it insures the exposure exactly rather than its average, and is bought where one large exposure justifies the bespoke cost. Banks also transfer CVA risk in portfolio form, through credit-linked notes and guarantees sold to investors such as insurers and credit funds that want the spread and can carry the risk. And what a desk hedges follows the capital charge as well as the price: SA-CVA recognizes hedges of the exposure's rate and FX sensitivities and BA-CVA does not (Section 27.6), so a desk under SA-CVA hedges the market side of CVA as well as the credit side.

FVA and MVA are hedged, to the extent they are, by managing the firm's own funding profile; in most banks the treasury charges them to the desks through funds-transfer pricing, at an internal rate for the term and liquidity of the funding each trade consumes. KVA is generally not hedged at all; it is managed by not doing the trades.

Why the XVA desk loses money in a crisis

An XVA desk that has hedged carefully can still report large losses, for three recurring reasons.

Proxy basis. When a specific counterparty's spread widens while the index used to hedge it does not, the CVA rises and the hedge does not respond. In a broad credit sell-off the proxy works; in an idiosyncratic default it does not, and idiosyncratic defaults are exactly when CVA matters.

Cross-gamma. CVA depends on exposure *and* on credit, and the exposure depends on rates and FX, so a rate move changes CVA even with credit spreads unchanged, and a spread move changes how much a rate move matters. That interaction is CVA's **cross-gamma**. Hedging the credit sensitivity alone leaves that interaction unhedged, and the cross-terms are largest when both move together, which is the wrong-way risk of Section 9.9 as the hedging desk sees it.

Hedging costs. Buying CDS protection consumes capital and funding of its own, so a desk that hedges its CVA fully has converted a credit exposure into a capital and funding exposure, which is an improvement, not a cure.

The deeper issue is that CVA hedging is an attempt to hedge an option on a portfolio whose composition changes daily, written on a counterparty whose credit may not trade, in a market where correlations move when it matters. Doing it imperfectly is not a failure of execution; it is the nature of the problem.

9.11 THREE DIFFERENT NUMBERS

A recurring source of confusion inside banks is that the same trade carries three XVA figures that are all correct and do not agree.

The accounting number is what appears in published financial statements. It must satisfy fair value accounting standards (IFRS 13 internationally and ASC 820 in the United States), which require an exit price, what a market participant would pay, including the risk that either party does not perform, and allow a portfolio of trades with one counterparty to be valued

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net. That admits CVA, DVA, and now FVA, which any market participant would price, and generally excludes KVA, because a firm's own hurdle rate is not a market participant's.

The regulatory number is what drives capital. It includes the CVA capital charge of Section 9.7, applies prudential filters (removing DVA, as Section 9.3 described), and follows rules that may differ from the firm's own models.

The pricing number is what the desk charges the client. It includes everything the trade actually costs the firm, KVA included, because a trade that does not clear its capital hurdle destroys value regardless of how it is booked.

The client has an accounting number of its own, seen from the other side: its fair value carries CVA on the dealer and DVA on itself. Where it has designated the swap as a hedge of a loan, those adjustments have no counterpart in the hedged item, so they show up in its profit as hedge ineffectiveness.

The three diverge systematically, and the divergence is not an error to be reconciled away. A trade can be profitable on the accounting number, capital-destroying on the pricing number, and punitive on the regulatory one. Firms that manage to only one of the three make predictable mistakes: accounting alone produces trades that consume capital without earning it, regulatory alone an optimization of rules rather than of economics, and pricing alone published results that surprise the market. The practical resolution is that pricing decisions use the pricing number, published accounts use the accounting number, and the difference is explained rather than eliminated.

9.12 XVA AND THE MEANING OF VALUE

The chapter began with the claim that "the value of the swap" does not identify a number. What survives can be stated precisely. The cash flows on their own still have a well-defined value: the forward rates of Chapter 7, discounted on the collateral curve, a number two dealers will agree on to within a rounding error. What is not objective is the price, which depends on who the counterparty is, what the master agreement and CSA say, how the dealer funds itself, what capital the trade attracts, and what is already on the books between the two parties, so two dealers quoting the same client on the same trade may legitimately differ by tens of basis points, and neither is wrong. This is a genuine change in what a derivative *is*. Before the crisis, a swap was a bundle of cash flows and its value was a property of those cash flows. It is now more accurate to say that a swap is a bundle of cash flows plus a relationship, and that a substantial part of its price is the cost of maintaining the relationship. The framework that prices the relationship remains contested, for three reasons.

The components overlap and the overlaps are not agreed. Besides the FVA–DVA problem of Section 9.4, capital is held against counterparty risk that CVA has already priced, and MVA and FVA both price funding. Every firm decides for itself what to include, so the totals are not comparable across firms.

Some components are not market prices. CVA is disciplined by hedging: a dealer that charges too little can be arbitraged. KVA is not, for the reasons Section 9.7 gave, and calling it a valuation adjustment gives it a status it has not earned.

The precision is illusory. Simulations over years, credit spreads taken from thin markets, and unobservable correlations cannot support a total reported to the nearest thousand dollars, as Table 9.1 reports it.

None of this argues for ignoring the adjustments. The costs they represent are real, and a firm that does not charge them will discover them elsewhere. It argues for treating XVA as a disciplined framework for allocating known costs, rather than as a valuation in the sense that Chapters 13 and 15 use the word.

Counterparty credit adjustments predate the crisis: Sorensen and Bollier (1994) and Duffie

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and Huang (1996) already treated a swap's counterparty exposure as an option that is not symmetric between the two parties, and by the mid-2000s large dealers ran CVA desks. What came after 2008 was everything else, collateral-rate discounting, funding, margin and capital, and the works that introduced each are set out in the Further Reading.

SUMMARY

Exposure is $\max(V(t), 0)$: counterparty risk is an option on the portfolio's value struck at zero, so it needs a distribution rather than a point estimate, and it is measured across the netting set, where an offsetting trade attracts a lower charge. An interest rate swap's exposure humps early in its life, while a cross-currency swap's rises to maturity because the notional is exchanged then. Expected exposure serves pricing and potential future exposure serves credit limits, and much internal disagreement comes from not saying which is meant.

CVA is the market price of the counterparty's option to default, priced the way everything in this book is priced, by what the hedge costs: CDS protection on the counterparty, sized to the expected exposure. It therefore uses market-implied rather than historical default probabilities, because it is the cost of a hedge and not a forecast of a loss. DVA is its mirror image, required if two parties are to agree on a value, but it turns a deterioration in a firm's own credit into a reported gain, so regulators filtered it out of capital and it survives in accounts rather than in pricing.

Three further adjustments price funding and capital. FVA charges an uncollateralized position at the dealer's own funding spread; Hull and White argued that funding cost is not a property of the instrument, and practice settled the question through accounting recognition. MVA funds initial margin, so clearing and margin rules converted part of the charge for counterparty risk into a funding charge rather than removing it. KVA prices capital at the firm's hurdle rate and is the most strained of the set, since it embeds a management choice in an instrument's value.

Together the adjustments can come to many times the interdealer spread: the XVA is not a correction to the price but substantially is the price. Clearing reduces the total without removing it. ColVA prices how collateral is remunerated and chosen; thresholds and the margin period of risk appear as CVA and FVA on the exposure they leave.

CVA the price and the CVA capital charge are different objects, the second held against the volatility of the first, so reducing one need not reduce the other. Wrong-way risk makes the standard calculation a lower bound. And the same trade carries three defensible numbers, for accounting, for regulatory capital and for pricing, and they differ by design rather than by error. XVA is managed centrally because the quantities are portfolio properties, and hedged imperfectly because most counterparties have no traded credit. The framework remains contested, but the costs are real, and a firm that does not charge them meets them somewhere else.

FURTHER READING

Gregory's *The xVA Challenge* is the standard practitioner reference and Green's *XVA* treats the whole set of adjustments in one volume; Brigo, Morini, and Pallavicini's *Counterparty Credit Risk, Collateral and Funding* gives the derivations this chapter states, in full. Burgard and Kjaer (2011) is the source of the partial-differential-equation representation with bilateral credit and funding costs. The FVA argument is worth reading in the original: Hull and White's "The FVA Debate" and their later "Collateral and Credit Issues in Derivatives Pricing" set out one side, Laughton and Vaisbrot's 2012 reply in *Risk* the other, and Andersen, Duffie, and Song's "Funding Value Adjustments" the resolution in terms of who bears the cost. Piterbarg's "Funding Beyond Discounting" is the source of collateral-rate discounting, and Green, Kenyon,

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and Dennis's "KVA: Capital Valuation Adjustment by Replication" introduced the capital charge. Capriotti (2011) explains the adjoint algorithmic differentiation that makes a nightly exposure run affordable at all.

PRACTICE QUESTIONS

9.1 Explain why counterparty exposure is $\max(V(t), 0)$ rather than $|V(t)|$, and why this makes counterparty risk an option.

9.2 Explain why an interest rate swap's expected exposure peaks partway through its life while a cross-currency swap's rises to maturity, and state which carries more counterparty risk and why.

9.3 Explain why exposure is computed on the netting set rather than trade by trade, and why a dealer might quote a better price for a trade that offsets its existing position with a client.

9.4 A three-year uncollateralized swap has expected exposures of \$1.2m, \$1.5m, and \$0.9m at mid-year points. The counterparty's CDS trades at 180 basis points, recovery is 40%, and discount factors are 0.97, 0.93, and 0.89. Compute the CVA, first as the cost of insuring the exposure and then as loss given default times the default rate the spread implies, and confirm the two agree.

9.5 Explain why CVA is computed from the counterparty's CDS spread rather than from its historical default rate, and what would happen to a dealer that used the historical rate.

9.6 State the argument for DVA and the objection to it, explain why regulators removed DVA from capital, and state what that leaves banks managing.

9.7 Set out the Hull–White argument against FVA and the practitioner response. In what sense was the debate settled and in what sense was it not? Identify where FVA and DVA arguably double-count, and state what a firm would have to believe to include both.

9.8 Using the trade of Example 9.4, recompute MVA at a funding spread of 90 basis points and average initial margin of \$2.5 million, compare it with the CVA of Example 9.2, and explain why KVA is nonetheless often the largest adjustment on a long-dated uncollateralized trade.

9.9 A portfolio's value at the peak of its exposure profile is normally distributed about zero with a standard deviation of \$9 million. Compute the expected exposure and the 95% potential future exposure, and explain which is used for pricing, which for credit limits, and why.

9.10 A client requests a \$3 million threshold in its CSA. Using a CDS spread of 150 basis points, a funding spread of 70 basis points, an average uncollateralized exposure of \$1 million, and a discount-weighted duration of 4.0, estimate the cost of the concession. How would the funding cost change if the client were instead a sovereign that insisted on a one-way CSA, under which only the dealer posts collateral?

9.11 Distinguish CVA the valuation adjustment from the CVA capital charge, explain why the latter exists, and give one action that reduces both and one that hedges the price while leaving most of the charge in place.

9.12 An XVA desk reports a large loss in a quarter during which it was fully hedged on credit spreads. Give three distinct explanations consistent with the hedging having been done competently.

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FURTHER QUESTIONS

9.13 A dealer prices a ten-year uncollateralized swap for a corporate client at theoretical value plus CVA only, ignoring FVA, MVA, and KVA. Describe what happens to the firm over the following decade, identifying which cost appears where in its accounts.

9.14 Give three examples of wrong-way risk, one specific and two general, and explain in each case why insuring the average exposure fails. How would you adjust the calculation in each case?

9.15 Argue that KVA does not belong in a valuation at all, and then argue that omitting it guarantees that a dealer's derivatives business destroys shareholder value. Which treatment would you adopt, and how would you present it to a trading desk?

9.16 Post-crisis reform moved derivatives into clearing to reduce counterparty risk. Using Table 9.1, assess how far it succeeded, and explain precisely what was reduced and what was substituted for it.

9.17 Most of a dealer's counterparties have no traded CDS. Explain how CVA is hedged in practice, what basis risk this leaves, and when that basis risk would be most damaging. Relate your answer to the default correlation of Chapter 24.

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